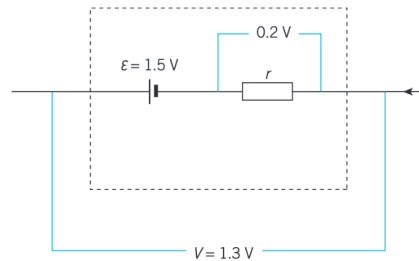


Internal Resistance

Lost Volts

When there is current in a power source, work is done by the charges as they move through the power source. Due to this, some energy is wasted when there is a current in the power source - not all energy transferred to the charge is available for the circuit.



In this instance, the p.d. measured at the terminals is less than the actual e.m.f. We call this difference **lost volts**. Here, the **lost volts** is 0.2 V because 0.2 volts have been lost to the resistor.

From K_2 , $\sum \epsilon = \sum V$ in a circuit. In normal use, e.m.f. does not change, but by changing the current the lost volts and terminal p.d. change too.

As current increases, lost volts increases and terminal p.d. decreases.

The e.m.f. is always more than the terminal p.d. (unless $I = 0$). When the current is very small, $\epsilon = V$.

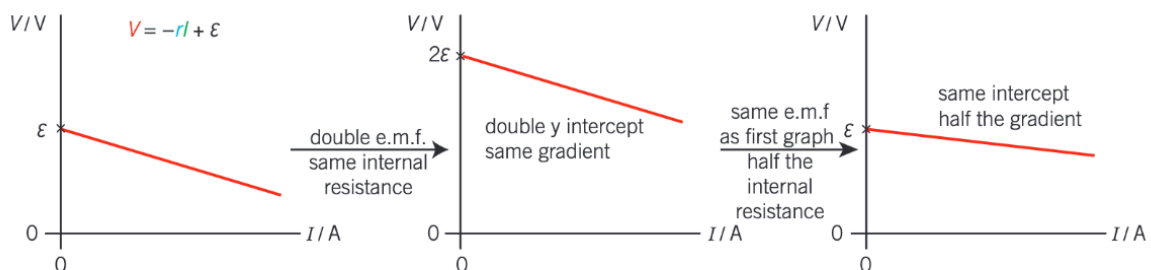
Some useful equations

$$\begin{aligned}\epsilon &= V + \text{lost volts} \\ &= V + Ir \quad \text{where } r \text{ is the internal resistance.} \\ &= IR + Ir \\ &= I(R + r)\end{aligned}$$

This relationship is essentially a version of $V = IR$ that takes into account the internal resistance of the power source.

Graphs of V against I

Different power sources have different e.m.f.s and internal resistances. However, graphs for different power sources will still follow the same general trend.



▲ **Figure 6** Changing the e.m.f. ϵ or the internal resistance r of the power source changes the graph

Here, we can see that changing the e.m.f. or internal resistance of the power source changes the graph.

Summary Questions

1. A rechargeable battery should have a low internal resistance, so that the energy used in charging the battery can be used by the components connected to the battery efficiently.

A high-voltage power supply should have a very high internal resistance, so that fatal shocks cannot be delivered by the power supply to a person.

2. As the current increases, the terminal p.d. decreases because more power is lost to the internal resistance of the battery.

3. a. $V_{\text{lost}} = Ir = 1.5 \times 2 = 3.0\text{V}$

- b. $V_{\text{terminal}} = IV - Ir = 13.5 - 3 = 10.5\text{V}$

- c.
$$\begin{aligned}\frac{E}{t} &= VI \\ &= 3 \times 1.5 \\ &= 4.5 \text{ J/s}\end{aligned}$$

- d.
$$\begin{aligned}R &= \frac{V}{I} \\ &= \frac{10.5}{1.5} \\ &= 7.0\Omega\end{aligned}$$

4. a.
$$\begin{aligned}R_{\text{total}} &= 50 + \left(\frac{1}{90} + \frac{1}{45}\right)^{-1} \\ &= 80\Omega\end{aligned}$$

- b. Terminal p.d.:

$$\begin{aligned}V &= IR \\ &= 0.1 \times 80 = 8.0\text{V}\end{aligned}$$

So lost volts is 4.0V.

- c.
$$\begin{aligned}r &= \frac{V_{\text{lost}}}{I} = \frac{4}{0.1} \\ &= 40\Omega\end{aligned}$$

5.

